

Small Project 1 - Tribal ML Benchmark & Population Optimization

Fundamentos de Aprendizagem Automática

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Small Project 1: Tribal Machine Learning & Population Optimization

1. Context & Objectives

Pedro Domingos' *The Master Algorithm* frames Machine Learning through five core tribes: 1. **Symbolists** (Rule induction, Decision Trees) 2. **Connectionists** (Multi-Layer Perceptrons, Neural Nets) 3. **Evolutionaries** (Genetic Algorithms, Differential Evolution) 4. **Bayesians** (Naive Bayes, Bayesian Networks) 5. **Analogizers** (k -NN, Support Vector Machines)

In this project, students will:

- * Benchmark implementations from the 5 tribes on complex non-linear classification and regression tasks.
- * Investigate the evolutionary tribe in depth: evaluate population-based optimization algorithms (Differential Evolution, Particle Swarm Optimization).
- * Implement and evaluate modern population initialization methods:
 - Uniform Random Initialization
 - Opposition-Based Learning (OBL)
 - Empty Space Search (ESS)
 - **OBLESA** (Opposition-Based Learning with Empty Space Search)
- * Perform knee detection (using the knee library) on performance trade-off curves.

2. Deliverables

1. **Source Code & Notebooks:** Clean, documented implementation conforming to ruff linting standards.
2. **Technical Report (PDF, max 6 pages):**
 - Empirical comparison of the 5 tribes.

- Exploration vs exploitation analysis of population optimizers.
- Benchmark comparing random init vs OBL vs ESS vs OBLESA across varying dimensions.
- Analysis of knee points in hyperparameter sensitivity.

3. Important Dates & Submission

- **Release:** Week 3 (October 01-02, 2026)
- **Submission Deadline:** Class 8 (November 05-06, 2026)
- **Weight:** 25% of final course grade.